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How Shared Gender Identity with Teaching Assistants Relates to Student Outcomes in an Undergraduate Algorithms Course

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Abstract

Background and Context. An ongoing thread in computing education research is how to increase women's participation in computing. One potential way is to improve their sense of belonging, as prior research has found that individuals' sense of belonging is related to their persistence in computing. Research from other STEM fields has shown that students benefit from sharing a gender identity with their professor. However, there is limited work on how a teaching assistant's (TA) identity relates to students' outcomes, especially in computing contexts.

Objectives. We studied how sharing a gender identity with their discussion TA relates to students' performance, sense of belonging, and comfort level participating in required discussion sections. Specifically, we investigated whether there were significant differences between male and female students in each of these outcomes and whether these differences were smaller when students shared a gender identity with their TA.

Method. Using data from two offerings of an upper-level algorithms course ($N = 298$ students), we compared outcomes across groups of students that did versus did not share a gender identity with their discussion TA.

Findings. In our context, we did not find significant evidence of a relationship between shared gender identity with TAs and student outcomes.

CCS Concepts

• **Social and professional topics → Computing education**.

Keywords

teaching assistants; demographics; performance; sense of belonging

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1 Introduction

Prior research in computing education has shown that women have a lower sense of belonging compared to men [3, 9, 15]. Since individuals' sense of belonging is positively correlated with their performance and persistence [9, 14], this lower sense of belonging likely contributes to the under-representation of women in computing [6, 17].

One possible approach to closing these gender gaps is to hire a more diverse teaching staff. Studies on student-professor interactions in other fields have shown that students are more likely to persist in their major and less likely to drop a course when they share a gender with their professor [2, 8]. However, because many computer science (CS) courses are large [17], most students likely have very limited interactions with their professors and interact more frequently with teaching assistants (TAs) [12]. Students' interactions with their professors likely have a different dynamic than their interactions with (near-peer) TAs [18]. As a result, we cannot assume that these benefits are generalizable to all teaching staff.

It is important to investigate how sharing a gender identity with a TA relates to students' experiences in a course. Research in other STEM fields found that sharing a gender identity with graduate TAs (GTAs) is not correlated with students' performance [7, 10]. However, to our knowledge, there are no works that explore how sharing a gender identity with GTAs relates to students' sense of belonging. Additionally, while existing research has found that undergraduate TAs can positively impact student performance [5, 13] and sense of belonging [12, 18] in computer science classrooms, there are no works that specifically examine how shared gender identity relates to these outcomes.

Our study examined whether synchronous interactions with TAs that shared a gender identity related to increases in students' (**RQ1**) performance, (**RQ2**) sense of belonging, and (**RQ3**) comfort level participating in required discussion sections.

For each metric (RQ), we checked for gaps between male and female students and evaluated whether sharing a gender identity with a TA correlated with a narrowing of those gaps. We compared students' outcomes both within and across genders. Specifically, we 1) compared female students who shared a gender identity with their TA to female students who did not, and 2) compared male students to female students, first considering all students, then filtering for students who shared a gender identity with their TA.

We found that male students had significantly higher performance and were more comfortable participating in discussion than female students. However, we did not find significant evidence that sharing a gender identity with a TA was related to these differences. Moreover, we did not find significant evidence that a student's sense

of belonging was related to sharing a gender identity with their TA.

2 Related Work

2.1 How Shared Identity with GTAs Impacts Students' Performance

There is limited research on how shared identity with GTAs impacts students' performance. Griffith and Main [7] found that female students in an introductory engineering course were more likely to choose a higher-earning field within engineering, but not earn higher grades, if they were in class sections led by female GTAs. We note that there was one GTA for each 120-student section, so there were likely limited interactions between the students and the GTA. Similarly, Lee and Ing [10] found no significant difference in the academic performance of students from an introductory biology lab course who shared a gender (and race/ethnicity, low-income status, or first-generation status) with their GTAs.

Neither of the courses in these studies utilized undergraduate TAs (UTAs). Although Tari et al. [18] found that Asian women value UTAs with a shared gender and ethnicity, we do not know whether sharing a gender identity with a UTA relates to any student outcomes. Our study contributes to the literature by including UTAs in our analysis, having a higher TA:student ratio, and tracking specific TA:student interactions with attendance data from discussion sections.

2.2 Benefits of UTAs in Computing Courses

Prior research has found that UTAs can positively impact students' performance and sense of belonging in computing. Pivkina [13] found that students had higher average grades and pass rates in offerings of a data structures and algorithms course that used peer learning assistants. They also found higher pass rates in offerings of a discrete math course that used peer learning assistants. Similarly, Erdei *et al.* [5] found that in introductory programming courses, students who had UTAs in their lab section performed better on programming assignments than students who only engaged in pair programming during lab [5]. Perlmutter *et al.* [12] interviewed seven pairs of students and TAs that interacted in office hours and found that TAs can support students' sense of belonging by treating them kindly, helping them understand course material, and creating communities.

However, none of these studies considered the UTAs' gender identity and whether it relates to their impact on students. Our research begins to fill this gap in the literature.

3 Context

Our data is from the Fall 2024 and Spring 2025 offerings of an upper-level algorithms course at a medium-sized, research-oriented, private U.S. university with 15-week semesters. 340 of the 475 students enrolled across the two offerings (72%) consented to data usage. This course is required for the CS major, and students widely consider it to be one of the hardest courses in the major. At our institution, students declare their major at the end of their second year, and there is no restriction on the number of students in a given major. The mode student of this course was a third-year. The first author worked closely with both offerings as the course coordinator and was familiar with both the TA and student experience.

The weekly cadence of the course included two 75-minute lecture sections with all students and one 75-minute discussion sections with 20–30 students each. The discussion sections were led by either two UTAs or one UTA and one GTA. Within each section, no single UTA or GTA was given more control over the delivery of the course material (i.e. no one appeared to have more "seniority"). As a result, students were unlikely to be able to easily differentiate between a UTA and a GTA. Since we are studying student interactions with near-peer teaching staff, we treated UTAs and GTAs as the same in our analysis and refer to them collectively as TAs.

During the discussion sections, the TAs guided students through practice problems in a collaborative format. Students were asked to work on the problems in small groups with their classmates before the TAs discussed them with the entire section. The TAs also walked around the room and offered help to students while they were working in small groups. Attendance was taken during discussion and constituted 4.5% of students' course grade in Fall 2024 and 9% in Spring 2025.[1] Since students enrolled in their sections without knowing who their TAs were and TAs were assigned to sections based on their availability, the matching of a student to their TAs was effectively random.

In both offerings, the final exam was cumulative and constituted 35% (Fall 2024) and 30% (Spring 2025) of students' grades. We used students' final exam scores as a measure of their performance because we were interested in students' mastery of the course content. Since we combined two semesters of data, we normalized the exam scores by using percentiles in our analysis instead of raw scores. We did not analyze students' midterm exam scores because both offerings automatically dropped students' lowest midterm exam score when calculating final grades. Because of this policy, students may not have tried their best on all of their midterm exams, meaning their midterm exam scores may not reflect their mastery of course content. We did not use students' homework scores as a measure of their performance because both offerings used alternative grading methods that resulted in most students receiving full credit.

4 Methods

4.1 Data Collection and Coding

We sent students the same course survey in the first and last week of each offering. In the survey, we measured students' sense of belonging using 7-point Likert scale questions from Moudgalya *et al.*'s validated instrument on sense of belonging [11]. To reduce survey fatigue, we did not use all 30 items from the instrument. Rather, we chose the four questions that constituted the *Membership* factor of the instrument. We chose this factor because it had both the highest explainability of overall sense of belonging and the highest internal reliability (Cronbach's α). The instrument was validated in an introductory CS context, so we performed our own internal reliability test and obtained α values of 0.95 and 0.97 for the start and end surveys, respectively, which indicated very high reliability [4].

The surveys included two additional 7-point Likert scale questions: "I feel comfortable asking questions in discussion," and an

[1] In both offerings, students could skip 3 of the 12 discussions with no penalty.

attention check that asked students to select "Neither agree nor disagree." The seven students who selected a different option were left out of our analysis.

Students' responses to each Likert scale question were numerically coded from 1 (Strongly Disagree) to 7 (Strongly Agree). We used the mean of students' responses to the four Membership questions as a measure of their overall sense of belonging at the start and end of the semester. We used the difference in the means (calculated as end - start) as a measure of the students' *change in sense of belonging* during the course.

We collected students' demographic data using open-ended free response questions. Although a response was required, participants were explicitly told that they could type "Prefer not to respond" or another phrase with the same meaning. Students who wrote `Male, M, Man, Make, Cisgender Male, Mle` or `He/Him` were coded as `Male`; students who wrote `Female, Woman` or `F` were coded as `Female`; students who wrote `Non-Binary` or `Gender-Fluid` were coded as `Non-binary`; and students who wrote `Prefer not to respond` were coded as `Blank`. Although we explicitly asked students to describe their gender identity, students who responded with "Male" or "Female" outnumbered students who responded with "Man" or "Woman" at a 14:1 ratio. We therefore use "Male" and "Female" to describe students' gender, out of respect to the students' preferences.

In our analysis, we separated students into two groups ("Match" v.s. "No Match") based on whether they shared a gender identity with at least one of their discussion TAs. All recitations had at least one male TA, so there were zero male students that did not match on gender identity.

4.2 Data Filtering

Table 1 contains the demographic breakdown of students and TAs in our dataset after all filtering. We excluded students who identified as `Nonbinary` and `Blank` from our analysis. There were no TAs who identified as `Nonbinary` or `Blank` and therefore no TA for these students to match with. We also excluded students who attended fewer than 8 discussions from our analysis. We used a cutoff of 8 because there was a clear drop-off between students who attended 7 versus 8 discussions (see Figure 1). Additionally, students needed to attend at least 9 discussions for full credit. Students who attended 8 discussions were just shy of that requirement but still had a significant number of interactions with their TA. Lastly, for RQ2 and RQ3, we filtered out students who did not complete both surveys. This is because we investigated *changes* in students' sense of belonging and comfort participating in discussion.

This left us with data from $N = 298$ students for RQ1, and $N = 231$ students for RQ2 and RQ3.

4.3 Data Analysis

We used Mann-Whitney U tests to analyze our data because 1) the distribution of exam scores (used for RQ1) did not pass the Shapiro-Wilk test for normality ($p = .0084$) [16], and 2) RQ2 and RQ3 used ordinal Likert-scale data. For each outcome (performance, sense of belonging, comfort participating in discussion), we first compared male and female students before and after filtering for a gender match with a TA. Then, we compared female students who had a female TA with female students who did not. We applied a Holm-Bonferroni correction [1] for each group of tests run on the same outcome and report only the corrected p-values in Section 5. We used a Holm-Bonferroni correction as it reduces the likelihood of having Type II errors [1].

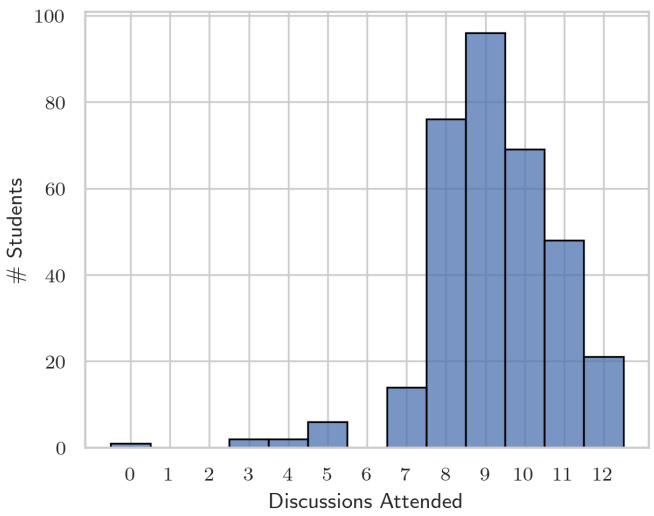

Figure 1: Distribution of Discussion Attendance (across both offerings)

Table 1: Distribution of Student/TA Gender Identities (across both offerings)

	# Students	*# TAs*
Male	176	24
Female	122	11
Non-binary	<5	0
Blank	23	0

5 Results

5.1 Performance

Our results suggest that shared gender identity with a TA is not related to students' exam performance. When we compared the exam scores of all male and female students, we found that female students had a lower median score (see Table 2). This trend persisted when we considered only students who shared a gender identity with their TA. Although all male students shared a gender identity with their TA, we still performed this test to see if the subset of female students who shared a gender identity with a TA would perform more similarly to male students. However, we found the same effect size (r) for these two tests, indicating that having a female TA did not close the performance gap.

When we explicitly compared female students that did and did not share a gender identity with their TA, we also did not find evidence that shared gender identity was related to exam performance. The result was not statistically significant, and there was only a

Table 2: Results of Mann-Whitney U Tests for exam scores, change in sense of belonging, and change comfort participating in discussion for male vs female students. r denotes effect size and Q1-Q3 denotes the first and third quartiles, respectively.

				Male			*Female*		
Outcome	**Group**	**p**	**r**	**n**	**Median**	Q_1-Q_3	**n**	**Median**	Q_1-Q_3
Exam Scores	All	**0.011**	0.18	176	74	60-84	122	65	60-75
	Match gender	**0.006**	0.18	176	74	60-84	74	65	56-75
Sense of Belonging Start	All	0.15	0.13	131	5.5	4.5-6.0	100	5.0	4.44-6.0
	Match gender	0.19	0.12	131	5.5	4.5-6.0	66	5.0	4.5-5.94
Sense of Belonging Change	All	0.53	−0.041	131	0	-0.50-1.0	100	0.13	-0.31-1.0
	Match gender	0.40	−0.091	131	0	-0.50-1.0	66	0.50	0-1.0
Discussion Start	All	**0.0036**	0.20	131	6.0	5.0-6.0	100	5.0	4.75-6.0
	Match gender	**0.031**	0.16	131	6.0	5.0-6.0	66	5.0	5.0-6.0
Discussion Change	All	**0.039**	−0.16	131	0	0-1.0	100	0.50	0-1.0
	Match gender	0.082	−0.14	131	0	0-1.0	66	1.0	0-1.0

Table 3: Results of Mann-Whitney U Tests for exam scores, change in sense of belonging, and change in comfort participating in discussion for female students. r denotes effect size and Q1-Q3 denotes the first and third quartiles, respectively.

			Match			*No Match*		
Outcome	**p**	**r**	**n**	**Med**	$Q1$-$Q3$	**n**	**Med**	$Q1$-$Q3$
Exam Scores	0.65	−0.041	74	65	56-75	48	65	60-74
Sense of Belonging Start	0.82	0.023	66	5.0	4.5-5.94	34	5.0	4.31-6.0
Sense of Belonging Change	0.36	0.15	66	0.50	0-1.0	34	0	-0.69-0.69
Discussion Start	0.19	0.13	66	5.0	5.0-6.0	34	5.0	4.0-6.0
Discussion Change	0.88	−0.015	66	1.0	0-1.0	34	0	0-1.0

Table 4: Count of students with negative/neutral/positive changes in scores for sense of belonging and comfort participating in discussion.

	Male			*Female*		
Outcome	**-**	**0**	**+**	**-**	**0**	**+**
Sense of Belonging	32	55	44	21	44	35
Discussion	32	51	48	10	40	50

slight negative effect (see Table 3) that is likely noise and not due to actual differences between the two groups.

5.2 Sense of Belonging

Our results suggest that shared gender identity with a TA is not related to changes in students' sense of belonging throughout a course. There were no significant differences between groups when we compared their sense of belonging scores at the start of the semester (see Tables 2 and 3). Figure 2 shows the distributions of scores for male and female students at the start and end of the semester. Table 4 contains the distributions of positive, negative, or no change for male and female students.

We found no significant differences both before and after filtering for students who shared a gender identity with their TA (see Table 2). Although female students who had a female TA had slightly higher changes, this difference was not statistically significant (see Table 3).

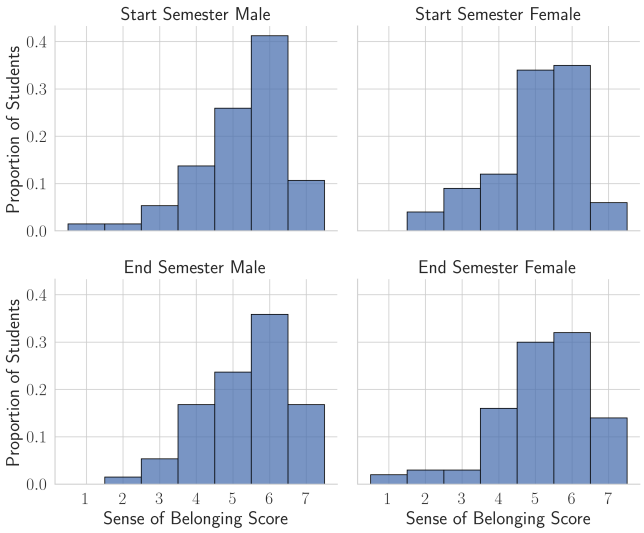

Figure 2: Sense of Belonging Scores for Male versus Female Students

5.3 Comfort Participating in Discussion

Although female students had higher increases than male students in their comfort participating in discussion, this effect did not appear to be related to shared gender identity with TAs. When we

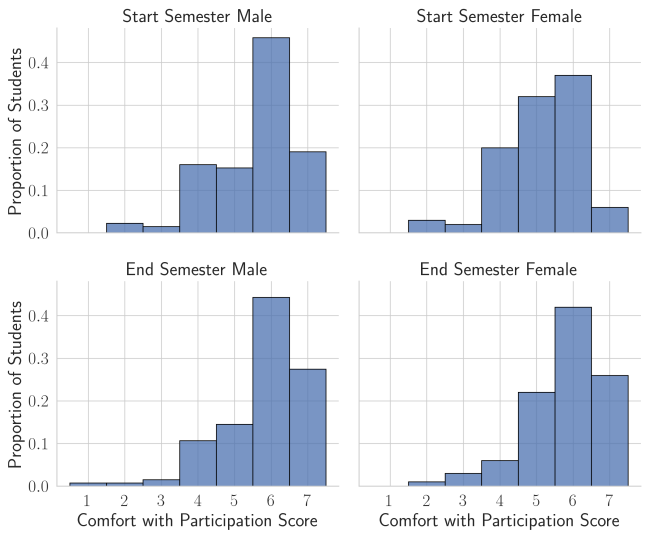

Figure 3: Comfort Participating in Discussion for Male versus Female Students

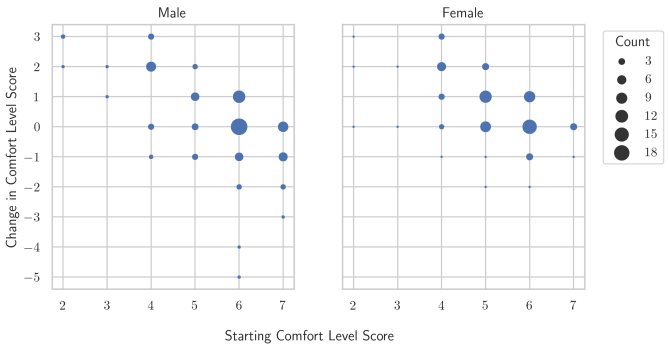

Figure 4: Change in Comfort Participating in Discussion for Male versus Female Students

considered *all* students (i.e. irrespective of a gender match), we found that male and female students had significantly different changes in their comfort participating in discussion (see Table 2). Upon entering the course, female students were less comfortable participating in discussion than male students. In Figure 4 and Table 4, we see that at the end of the semester, female students were generally equally or more comfortable than they were at the start of the semester. In contrast, there was a higher proportion of male students who had a negative change in comfort level. One explanation for this observation is that there was less "room" for male students to report a higher value because they started with greater comfort levels.

When we filtered for only students who shared a gender identity with their TA, we found no significant differences between male and female students' changes in comfort participating in discussion. Furthermore, we found that female students who shared a gender identity with their TA did not have significantly different changes in their comfort levels than female students who did not share an identity (see Table 3). This means that when female students were in a discussion led by a female TA, their comfort participating in discussion did not increase relative to male students nor female students who did not have a female TA. It should be noted that this also means that female students' comfort level participating in discussion did not *decrease* when they did not have a female TA. Collectively, these results indicate that female students' comfort level with participating increased in the environment of our discussion sections, and the gender of the TA leading these sections did not have a significant role in this increase.

6 Discussion

Our results indicate that students in our context do not perform differently or have different changes in sense of belonging and comfort participating in discussion when they share a gender identity with their TA.

One explanation for this observation is that students in higher-level courses are less sensitive to the effects (if any) of sharing a gender identity with course staff. The majority of students taking the course have already completed a minimum of three other core CS courses, in addition to CS electives. As a result, they may have a more inelastic sense of belonging informed by their previous experiences. Moreover, students who are affected by the lack of shared gender identity may have become accustomed to not seeing TAs like them in earlier courses, or they may have already dropped the major and are not present in our dataset.

Alternatively, the students may simply be operating in an academic environment that supports them enough such that having a TA with the same gender is not impactful. As we can see in Figures 2 and 3, the majority of our students are entering the course with neutral or positive responses to the questions on sense of belonging and comfort participating in discussion. Our institution is medium-sized and does not have a limit on the number of students in each major, so there is inherently less competition among students. Students also see some of their same peers throughout their coursework as many of our students follow the same sequence of classes. Also, all of our large courses (100+ students) are led by at least one instructor and a full-time teaching support staff, so there is a strong network to support students.

Furthermore, it is possible that students may simply not feel the salience of a shared gender identity with a TA. The positive effects of sharing a gender identity with an instructor may arise from the fact that instructors serve as a more distant role model, which a near-peer cannot be. Alternatively, students may notice their classmates' identities more than their TAs', or just not notice the identity of others in general. Other TA characteristics, such as their personality and quality of teaching, may be more noticeable and have a greater impact on the student experience. Our institution has a robust TA program, and this course prioritizes evaluating soft skills in the TA hiring process, conducts weekly training sessions for TAs, and provides TAs with feedback from both students and more experienced teaching staff. This process could create a strong TA team whose identities are not as salient to students.

7 Limitations and Future Work

Our data comes from an algorithms course taught at a research-oriented university in the US, so our results may not be representative of other instructional contexts. Additionally, we had no discussion sections with only female TAs due to scheduling constraints and the small number of female TAs. Moreover, as stated in Section 6, it is possible that the effects of shared gender identity are less pronounced for students in higher-level courses. Future work should examine these effects in introductory computing courses and at a diverse range of institutions.

Another limitation of our work is that our analysis did not consider shared racial identity with TAs. We had very few Black/African American and Latine/Hispanic TAs and students. After separating students by match status, we did not have large enough subsets of data to have any meaningful statistical power. Furthermore, our surveys collected demographic data using free-response boxes to give students greater freedom in expressing their identity. Many students were not specific in their responses and wrote in broader descriptors like "Asian." This means that, had we included race in our analysis, we would have coded some students who may not have felt a salient match with their TAs as matching, e.g. a South Asian student would be coded as matching with an East Asian TA. For future work, we recommend giving respondents more concrete identities to choose from in addition to the free-response text box.

Lastly, our work is limited in that we chose to exclude office hours interactions from our analysis. Most students in our dataset attended office hours fewer than five times. Additionally, our department's queuing system for office hours does not allow for customized questions. We believe that a meaningful analysis of office hours data would require many more interactions or qualitative data about students' experiences in office hours. Additionally, it is important to note that office hours interactions differ from discussion interactions in that students have much more freedom to choose which shifts to attend, as opposed to being (effectively) randomly assigned to a discussion.

Future work considering office hours could include short free response questions in either the queueing system or the course surveys that ask students about how they choose which shift to attend. Additionally, researchers could ask students to indicate whether there are TAs that they prefer to approach for help, or if there are any TAs in the course that they both interact with and feel like they can identify with. Lastly, interviews can be used to investigate how students and TAs think about interacting with others with a shared gender identity. This would help resolve some limitations with quantitative data that assumes that a shared identity is salient.

8 Conclusion

To our knowledge, our study is the first to examine the relationship between students' outcomes and whether they share a gender identity with a TA. Although prior work established that TAs can positively affect students' performance and sense of belonging, they treated each student-TA interaction as indistinguishable. We add to the literature by explicitly considering differing effects of social identities on interactions between TAs and students. We did not find significant evidence that a shared gender identity relates to differences in student outcomes. However, our study is only a first step towards understanding the phenomenon of shared identity. Additional research, particularly qualitative work, is needed to understand this relationship in other contexts. This would give instructors more information on what to prioritize when staffing a course and how the course staff can better support all students.

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